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Predicting How People Update Beliefs: Comparison of Individual-, Subgroup-, and Population-Level Dynamic Bayesian Networks

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Abstract

Modelling cognitive processes such as belief updating requires a decision about the level at which a model is fitted: a single model estimated from the whole sample obscures systematic differences between individuals, whereas individual-specific models yield noisy parameter estimates that hardly generalise beyond the observed participants. Clustering individuals with similar patterns of belief updating across individuals, including when, how, and under which conditions they revise their beliefs and fitting one model per cluster offers a compromise, yet its predictive value has received little systematic attention. In this paper, we cluster individuals in dynamic Bayesian networks (DBNs) that share a known and identical graph structure, so that clustering reduces to the conditional probability table (CPT) space, and we apply hierarchical agglomerative clustering to participants' CPTs, which yields a hierarchy of nested groupings and thus does not require the number of groups to be fixed in advance. Using a simulated belief updating study based on a cooperative gaze-following paradigm, we compare predictions for unseen trials obtained from (i) individual-specific DBNs, (ii) cluster-specific DBNs estimated from the identified groups, and (iii) a single DBN estimated from the entire sample. We evaluate them with classical measures of predictive accuracy and with theory-guided characteristics of belief updating. Cluster-based DBNs predict new trials about as accurately as individual-specific DBNs and significantly better than sample-level models. Thus, they recover interpretable subgroup-specific patterns of belief updating while retaining the generalisability needed to make predictions for previously unseen individuals.

1. Introduction

Understanding others requires people to continuously integrate new social information with their existing expectations. The extent to which individuals revise their beliefs in response to new evidence is therefore a fundamental process underlying social perception and interpersonal decision-making.

In general, perception is shaped through an interaction between bottom-up sensory signals and top-down influences, including prior assumptions about individuals or situations (Kok et al., 2013; McGovern & Otten, 2024). Mental models of other people are constructed in order to anticipate future states, such as the behaviour of other people. These models can be revised when new social information becomes available (Mende-Siedlecki et al., 2013). This process, commonly described as belief updating, entails adjusting prior judgements when confronted with evidence that may con-

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tradict earlier beliefs. The time required for such revisions depends on both the type of prior knowledge and the nature of the new evidence (Ferguson et al., 2019; Kammrath et al., 2007). People vary in the way they combine incoming information with existing beliefs, which results in differences in when they preserve or how quickly they revise their impressions of others (Cone et al., 2017).

People also construct internal representations of others' cooperativeness during interactions, integrating both personal and behavioural cues of the other person as well as the situational context (Rossetti et al., 2022). Such cooperative person models can lead to biased interpretation. For example, ambiguous information can be understood in ways that fit prior expectations, or contradictory evidence can result in biased updating of beliefs or mental models.

Understanding inter-individual differences in belief updating is important because people vary in how they integrate prior expectations about others with new observations or otherwise obtained information. These differences influence social sensitivity, adaptability, and decision-making. Understanding belief updating helps to explain the mechanisms underlying social cognition and can improve cooperation between humans and machines. It is particularly relevant for designing effective support in contexts such as human-robot interaction, where individuals may need help revising inaccurate beliefs. By predicting who is likely to resist or accept new information, researchers and practitioners can better tailor communication strategies. To be able to make such predictions, it is important to consider the trade-off between inter-individual differences that should be taken into account while avoiding overfitting and ensuring generalisability to new individuals.

Bayesian networks (BNs) offer a structured way to represent probabilistic dependencies between variables (Pearl, 1988). In practice, estimating the parameters of a BN from data introduces a trade-off that is particularly relevant for predictive applications: a single model fitted for the entire sample tends to obscure systematic differences between individuals, thereby reducing the individual fit and limiting the predictive precision of the model, while estimating a separate BN for each individual might result in noisy parameter estimates that are hardly generalisable. One way to address this issue is to partition individuals into clusters for which distinct BNs are learned. This can enhance predictive performance by allowing models to reflect group-specific patterns while still ensuring reliable estimation. Although BNs are widely used in probabilistic modelling, comparatively little attention has been paid to principled methods for clustering with fixed network structure, particularly with focus on using these BNs to make predictions for new data.

Clustering in probabilistic graphical models (such as BNs) has mainly been studied from two perspectives. First, *lifted probabilistic inference* leverages indiscernibility within a single model for compression and faster inference (Niepert & Van den Broeck, 2014; Poole, 2003). The state of the art to compute groups of variables for compression is the Advanced Colour Passing (ACP) algorithm (Luttermann et al., 2024), which has been extended to Hierarchical Advanced Colour Passing (HACP) (Speller et al., 2025) to build a hierarchy of lifted representations by succinctly increasing a tolerance parameter ε used to determine similarity (Luttermann et al., 2025). Second, on a population level, mixtures of BNs (Thiesson et al., 1998) use a class variable to switch between models. Furthermore, Bayesian hierarchical clustering (Heller & Ghahramani, 2005) groups individuals based on a model-comparison criterion (which, however, relies on generative assumptions). Both mixtures of BNs and Bayesian hierarchical clustering ignore the structural prior knowledge

available in our setting, where every individual shares the same known graph structure and only the conditional probability tables (CPTs) differ, so that clustering reduces to the CPT space.

A related alternative is provided by hierarchical (multilevel) models, which jointly model variation at different levels and allow individual- or group-level parameters to be combined (Gelman & Hill, 2006). Such models are particularly useful when individual differences are represented as variation around a population-level distribution and individual-specific parameters can be informed by observations from the respective individuals. However, for a previously unseen individual for whom no such observations are available, the individual-specific component cannot be estimated from the target individual's data, and predictions instead rely on the population-level distribution. Our clustering approach provides a different way of representing heterogeneity: when characteristics available prior to the prediction task allow a new individual to be associated with one of the identified clusters, the corresponding cluster-specific dynamic Bayesian network (DBN) can be used for predictions without requiring observations from that individual. This may be useful when individual differences are better represented by distinct subgroups rather than by continuous variation around a common population distribution. Moreover, our setting differs from conventional multilevel regression in that the model is a DBN, where conditional probability tables define dependencies within and across successive trials and predictions are recursively propagated over time.

Although there are many well-developed modelling algorithms and clustering approaches that are already in use, there is still a lack of systematic testing to determine whether they can be usefully applied to the analysis of cognitive processes under realistic conditions. Applying mathematically derived methods directly to real data, where the underlying structures are unknown and which, in studies of psychological or cognitive processes, is often noisy and influenced by many factors, can lead to misinterpretations and incorrect conclusions. In particular, the predictive accuracy of cluster-based models — as a compromise between individuality and generalisability — has not been sufficiently investigated and requires testing on simulated data in order to quantify their accuracy and thus ensure that the quality meets expectations, whilst also identifying any potential problems or difficulties in their use for modelling cognitive processes such as belief updating.

In this paper, we cluster individuals in (dynamic) BNs with identical known graph structures and investigate the predictive performance of DBNs fitted at different levels of aggregation: (i) individual-specific DBNs, (ii) cluster-specific DBNs, and (iii) a single DBN estimated from the entire sample (i.e., representing the population level). To derive cluster-specific models, we apply hierarchical agglomerative clustering to individuals' CPTs while assuming an identical known graph structure across participants. As a real-world inspired application, we consider a simulated belief updating study based on a currently planned cooperative gaze-following paradigm. The simulation generates groups of individuals whose underlying CPTs, i.e., cognitive processes of belief updating, differ regarding their indistinguishability, allowing us to assess if cluster-based DBNs can improve predictive accuracy relative to population-level models while simultaneously allowing generalisability to new data compared to individual-level models.

Specifically, our contributions are as follows. First, we derive cluster-specific DBNs that capture shared probabilistic dependencies, i.e., similarities in cognitive processes such as belief updating, among groups of individuals while maintaining greater generalisability than individual-specific models. Second, we use standard measures to investigate the predictive performance of DBNs esti-

mated at different levels of aggregation, namely individual-specific, cluster-based, and population-level models. This allows us to assess the trade-off between capturing individual differences and achieving generalisable predictions. We empirically show that the prediction accuracy of cluster-based DBNs performs approximately as good as individual-specific DBNs and much better than sample-level models in order to predict patterns in belief updating of individuals. Finally, we assess additional real-world oriented measures of prediction quality to show that the cluster-based DBNs can keep up with individual-specific DBNs (and both outperform sample-level models) in capturing underlying patterns of cognitive processes (here, belief updating) when predicting new data.

2. Background

BNs provide a principled way to deal with uncertainty, e.g., to model uncertain updating processes such as belief updating. A BN factorises a full joint probability distribution over a set of variables into a product of CPTs by combining a directed acyclic graph — whose nodes are the variables and whose edges encode direct probabilistic dependencies — with a CPT at each node that specifies the distribution of the node given the values of its parents (Pearl, 1988). A DBN extends a BN to time-evolving processes by chaining one BN per discrete time step, called a time slice, and adding inter-slice edges that encode how variables evolve from one slice to the next (Murphy, 2002).

Definition 1 (Dynamic Bayesian Network)**.** A *DBN* factorises a joint probability distribution over a sequence $\mathbf{X}_{0:T} = (\mathbf{X}_0, \ldots, \mathbf{X}_T)$ of random-variable vectors $\mathbf{X}_t = (X_t^{(1)}, \ldots, X_t^{(n)})$ via a pair $\mathcal{D} = (\mathcal{B}_0, \mathcal{B}_{\rightarrow})$ of *BNs*, each consisting of a directed acyclic graph over its variables together with a conditional distribution $\theta_{X|\mathrm{Pa}(X)}$ for every node X given its parents $\mathrm{Pa}(X)$. The prior $\mathcal{B}_0$ specifies the initial distribution $P(\mathbf{X}_0)$, whereas the transition $\mathcal{B}_{\rightarrow}$ is defined on $\mathbf{X}_{t-1} \cup \mathbf{X}_t$ with $\mathrm{Pa}(X_t^{(i)}) \subseteq \mathbf{X}_{t-1} \cup \mathbf{X}_t$ and carries conditional distributions only for the current slice, giving the transition model

$$P(\mathbf{X}_t \mid \mathbf{X}_{t-1}) = \prod_{i=1}^{n} P(X_t^{(i)} \mid \mathrm{Pa}(X_t^{(i)})).$$

Under the first-order Markov property and a stationary $\mathcal{B}_{\rightarrow}$, the joint distribution factorises as $P(\mathbf{X}_{0:T}) = P(\mathbf{X}_0) \prod_{t=1}^{T} P(\mathbf{X}_t \mid \mathbf{X}_{t-1})$.

Figure 1 shows the graph of an example DBN. Each node represents a variable, solid arrows define within-slice edges of a single time slice, and dashed arrows give the inter-slice dependencies between consecutive time slices. We next explain the meaning of the variables in more detail.

3. Simulating Cognitive Processes of Belief Updating

For data simulation, we closely follow a currently ongoing experiment examining belief updating in a cooperative context based on established cognitive processes. As data simulation is not our main focus, we only describe the most important experimental aspects and data structure required for the subsequent analysis. In our experiment, two people sit opposite each other and have a shared visual space with various objects of different values. The task of person A (observer) is to estimate the gaze of person B (initiator). The initiator looks at one of the objects and the observer has to guess which object the initiator looks at and communicates this choice accordingly. Afterwards, the observer

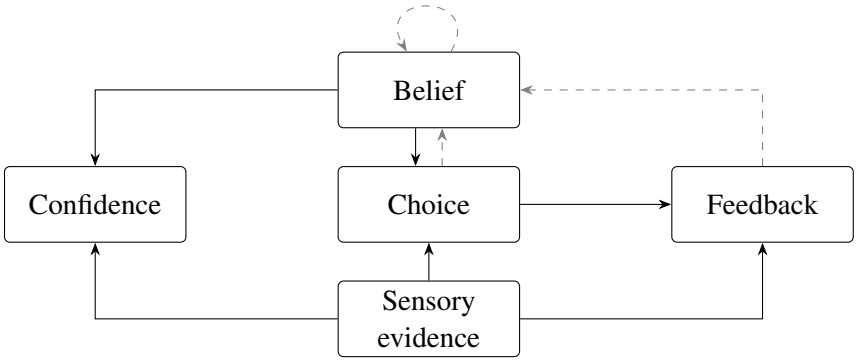


Figure 1. The DBN of the simulated data is shown with rectangles as variables and arrows as causal influences divided into solid lines for influences within the same trial (t) and dashed lines for influences on variables in the next trial ($t + 1$).

gets feedback from the initiator on whether or not this choice was correct. For each correct choice, both get points as a team — otherwise none of them get points, creating a cooperative setting. The points they get depend on the value of the selected object which can be high, medium, or low. Since the value of each object is known in advance, it makes most sense for the initiator to choose a high-value object (because it brings most points to the team), which is why we also assume the observer to have this initial assumption (of the initiator being cooperative and thus choosing mostly high-value objects). However, since we replace the initiator by a confederate (i.e., a person that only pretends to be a participant while actually being informed about the entire experiment), the initiator follows a predetermined order of objects, and we can thereby influence the degree of cooperativeness of the initiator. There are multiple trials, so the cooperativeness of the initiator can be varied and the observer's belief about the initiator's cooperativeness can be updated over time.

In short: We assume person A (observer) holds a *belief* regarding the cooperativeness of person B (initiator). Based on this belief and noisy, i.e., uncertain *sensory evidence* (gaze of person B), person A makes a *choice* and reports a *confidence* rating for the decision. *Feedback* is defined by the agreement of choice and evidence. The belief for the next trial is updated as a function of the previous belief, the choice, and the received feedback. All relationships between variables are assumed to be causal and are represented as CPTs. Each variable depends solely on its parents (which may belong to the same or the previous time step), indicated by incoming edges in Fig. 1.

Cooperativeness is staggered in three stages which is reflected in (expectation of) behaviour (here, selection of objects of three different values: high, medium, and low). Sensory evidence is set in line with the study, which is designed in blocks of trials. In most trials, the initiator (who is a confederate with fixed behaviour) shows high cooperativeness, but there is some noise, i.e., sometimes medium- or low-valued objects are selected, although they bring less points. In some blocks, the behaviour changes. There are *incongruent* blocks, where the behaviour of person B shows an increase in selecting low-value objects. In *random* blocks, the behaviour switches to a random selection of differently-valued objects. Afterwards, the behaviour changes back to primarily *congruent* (i.e., cooperative) behaviour of selecting high-valued objects.

Feedback is deterministic (*correct* if the choice matches the noisy sensory evidence, otherwise *incorrect*). The CPTs governing the remaining three variables (belief, choice, confidence) encode probabilistic associations and may vary across participants. Choice and confidence are updated probabilistically based on the congruency of belief and sensory evidence. In the corresponding DBN from Fig. 1, each time slice corresponds to one trial of the experiment. The within-slice CPTs capture dependencies among all variables within trial t, while the inter-slice CPTs model the transition of beliefs from trial t to trial $t + 1$.

Despite individual variability, we assume that participants can be meaningfully partitioned into distinct groups when it comes to belief updating. We distinguish four groups: (1) *Choice-based* updaters, who adjust beliefs more strongly following correct choices (particularly for small updates) but rarely revise beliefs after negative feedback (incorrect). (2) *Insecure* updaters, who exhibit a similar tendency but display higher stochasticity in their behaviour, suggesting weaker internal structure and more frequent switching between options. (3) *Sceptical* individuals, who tend to update beliefs more conservatively in a positive direction (e.g., towards increased cooperativeness). (4) *Persistent* individuals, who show minimal belief updating regardless of feedback or evidence.

Sampling is performed using the CPTs associated with the respective group assigned to each participant. Group membership is sampled with different prior probabilities, resulting in a structured but stochastic data generation process. Although using similar CPTs for participants within the same group, the exact data for the individuals will differ from each other due to the probabilistic associations which thereby automatically creates individual differences, even within groups.

This design aims to enhance simulation plausibility by incorporating realistic complexities such as proximity between groups and different group sizes, i.e., rare groups compared to more common ones which are more likely to dominate the overall sample. The intention is to narrow the gap between theoretical assumptions and empirical data, thereby improving the applicability of the results to real-world clustering problems. Although the simulated structure simplifies reality and cannot capture all aspects of empirical behaviour, it provides a sufficient basis for evaluating predictions based on clusters of individuals.

The R (R Core Team, 2026) implementation using *tidyverse* (Wickham et al., 2019) is available on GitHub: `https://github.com/R-oadrunner/Cluster-Predictions-in-BNs`

4. (Hierarchical) Clustering of Individuals

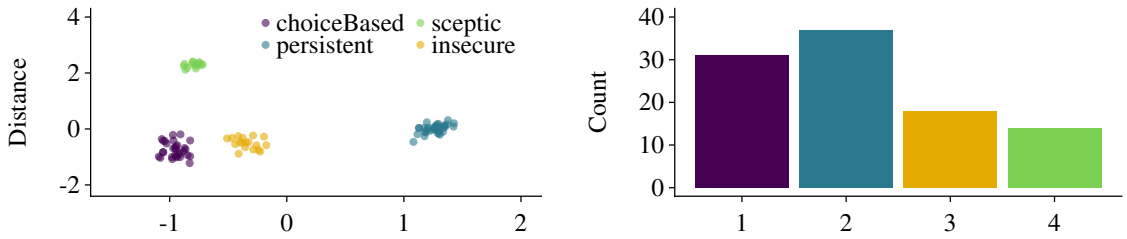
Separate models per participant best capture individual updating behaviour. However, individual models do not generalise well to the population with so far unseen individuals, whereas averaging across samples ignores meaningful differences. Clustering offers a compromise by grouping individuals with similar patterns while preserving differences between groups.

The clustering approach we apply is based on HACP (Speller et al., 2025), an agglomerative procedure originally designed to compress factor graphs but applicable to compute clusters in any collection of objects equipped with a distance. Starting from each object in its own group, HACP repeatedly merges the closest pair under the given distance and returns the full ordered sequence of distance values ε at which merges occur, yielding a hierarchy of progressively coarser, nested groupings (objects merged at some ε remain together for every larger ε). We transfer the procedure

underlying HACP to CPTs: applied to the pairwise distances between participants' CPTs, the same agglomerative sweep merges behaviourally similar individuals first, so that cutting the hierarchy at a chosen level (i.e., ε) produces groupings without defining the number of groups upfront.

To apply the hierarchical clustering to a simulated data set, we sampled a random data set of $N = 100$ participants with 1800 trials each, consisting of different blocks (congruent, incongruent, and random) as explained in Sec. 3. Each block has 60 trials each and is repeated 10 times. We apply the clustering and subsequent prediction first with a rather high number of trials to test the general accuracy since clustering is easier with more trials (because the groups are more homogenous and easier to distinguish from each other when there are more trials that can be used for calculating the CPTs). In Sec. 5.4, we will briefly evaluate the performance with less trials to ensure the validation also for data sets with less learning trials.

Fitting a DBN to each individual of the data set results in a list of CPTs for each individual. To calculate the distance between two individuals of the data set, we use the Root Mean Squared Error (RMSE) $= \sqrt{\frac{1}{L}\sum_{i=1}^{L}(x_i - y_i)^2}$ with x_i and y_i being the ith values of two vectors x and y of length L. To get a better understanding of the data set and its individuals, we can visualise these distances in a two-dimensional space, e.g., with multidimensional scaling using cmdscale in R (R Core Team, 2026), which can be seen in Fig. 2(a). The true groups of the sampled data set are highlighted accordingly (i.e., groups are given by the colours). As explained before, the groups of this example data set are rather easily separable due to the high number of trials. For less trials, the mean distance between individuals within the same groups would increase and the different groups might slightly overlap. However, as discussed in Sec. 5.4, the results presented in the following parts remain largely consistent even when the number of trials is reduced substantially.



(a) (b)

Figure 2. All four true groups of belief updating are shown in both figures with colours indicating the different true groups. (a) visualises distances between the individuals in a two-dimensional space and (b) the number of individuals (y-axis) in each cluster (1-4 on the x-axis) in relation to the true groups meaning single-coloured bars imply that the cluster is homogenous and only includes individuals belonging to the same true group.

We use the distance matrix with RMSE distances to apply the step-wise hierarchical clustering as described before. We visualise the increase in the distance values ε in Fig. 3 for all clustering steps. A slow, steady rise is visible until shortly before the end, where the increase suddenly gets a lot bigger. In the last step, all individuals are in the same cluster. A greater increase indicates a larger distance between the individuals or groups that are clustered in this step. We therefore stop

the clustering at the clearly visible point where ε increases largely (highlighted as a green point in the plot) which leads us to four groups — which is exactly the number of true groups in the data. The distribution of the true groups within the computed clusters is shown in Fig. 2(b). We can see that all clusters were perfectly identified, i.e., all groups are homogeneous and consist only of individuals who belong to the same true group.

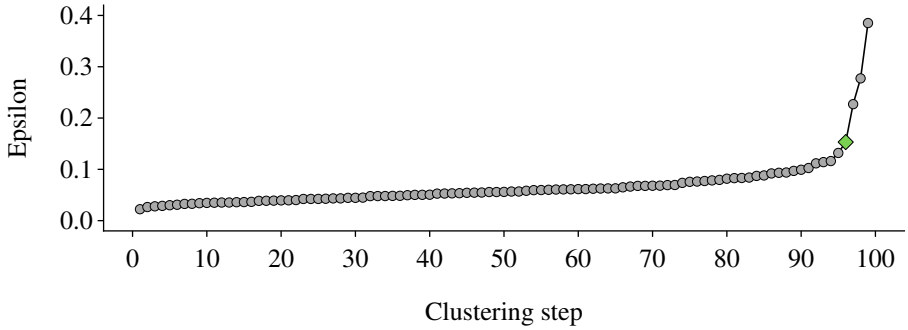


Figure 3. Increase in ε (y-axis) is plotted in dependence of the clustering step (x-axis). The green point indicates the optimal number of clusters.

5. Predicting Belief Updating in New Trials

For testing the prediction accuracy of the models, we use 180 new trials (i.e., much less than for learning) with a varied order compared to those used for fitting the DBNs (Scutari, 2010). For evaluation, we consider three different levels of aggregation when fitting DBNs for predicting individual belief updating. Specifically, we use (1) the individual CPTs fitted specifically for a single person, (2) the CPTs fitted based on all individuals within one identified cluster, and (3) CPTs when using the entire sample for learning. In (1), each participant uses different CPTs for prediction. In (2), there are four different sets of CPTs (one per cluster) that are used for all individuals in the corresponding cluster and in (3), all predictions are made using the identical sample-level CPTs.

For all three scenarios, the procedure is identical. The belief at the first trial is given (due to the experimental design, we assume all participants initially have the belief that the other person will be cooperative and thus select a high-value object). The fitted CPTs are then used to predict the probability of the states of the variables *choice* and *confidence* (although confidence is not essential here). These probabilities for the possible states are used to sample a state. Feedback results directly from this sampled choice and the sensory evidence (which is identical for all participants). Then, the probabilities of the possible states of belief are predicted using the sampled choice, the feedback and the belief from the previous trial. A state for belief for the next trial is sampled based on these predicted probabilities. This continues for all 180 trials. No observation of the variables despite the initial belief at the first trial and the sensory evidence is used for these predictions, i.e., predictions of the states at trial 180 are based on noisy probabilistic and potentially wrong sampled previous states. Since this whole procedure involves many probabilistic steps, the results differ for every run.

Therefore, we repeat this process of predicting one set of trials with a certain DBN 1000 times. From these 1000 iterations, we calculate the mean expectation $E[B_t] = \sum_i b_i p_i$ of the belief for each trial B_t by multiplying each possible state of belief b_i — 1 (low), 2 (medium), and 3 (high) — by their relative frequency p_i in that trial across all 1000 iterations. As measure of variation, we additionally calculate the standard deviation (SD) as $SD[B_t] = \sqrt{\sum_i p_i \left(b_i - E[B_t]\right)^2}$.

Examples of all three different model predictions compared to real data are presented in Fig. 4. By comparing the mean estimate of each trial (red lines) with the SD (red area) to actual participant data (black and grey lines), the accuracy of the models becomes directly visible. In the individual plot on top, there is only one participant whereas the plot in the middle shows all participants of that group (displayed as more transparent lines) and their mean estimate. Both plots have some visual similarities because the individual participant presented on top of Fig. 4 is also a member of the cluster represented in the middle. This cluster consists of members of the true group that update beliefs mostly in response to the previous choice as explained in Sec. 3. However, not all clusters have such obvious pattern — neither in the individual DBN predictions nor in those from the cluster — as we will see in Sec. 5.1. The last plot at the bottom of Fig. 4 includes all participants of the whole sample. There are some similarities in the pattern as in the other two prediction figures but they are less strong and the mean estimate differs visibly from the pattern in the other two plots. The grey lines demonstrate the variance in individual's updating over the trials.

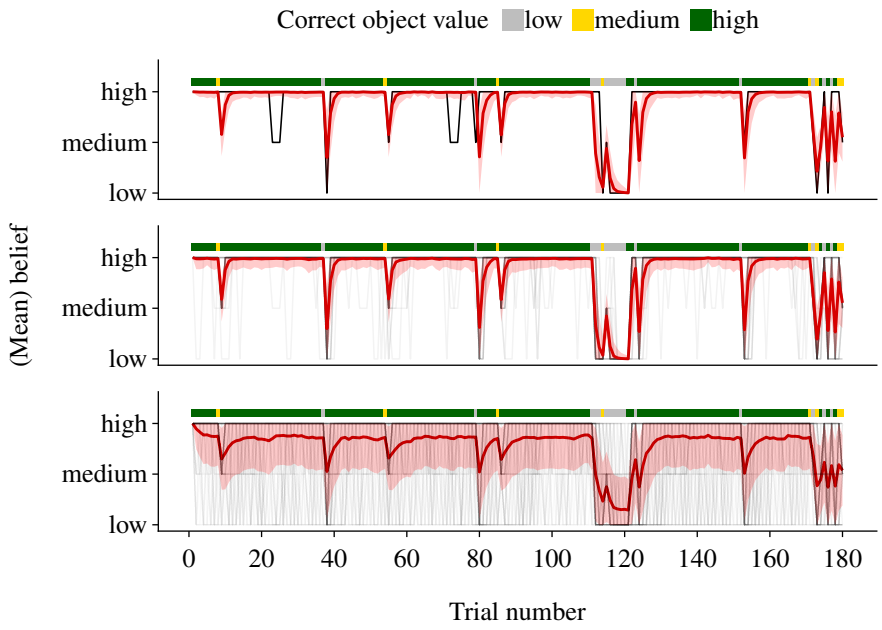


Figure 4. Mean predictions of the belief for each trial (red lines) with SD (red area) and actual participant data (black lines) are shown for an example individual-specific (top), cluster-based (middle) and sample-level DBN (bottom) with coloured illustration of the evidence on top of each plot.

5.1 Predictions for Different Clusters

Fig. 5 shows the mean estimate of the model for each trial (red lines) with the SD (red area) in comparison to actual participant data (black and grey lines) for the clusters not shown in Fig. 4.

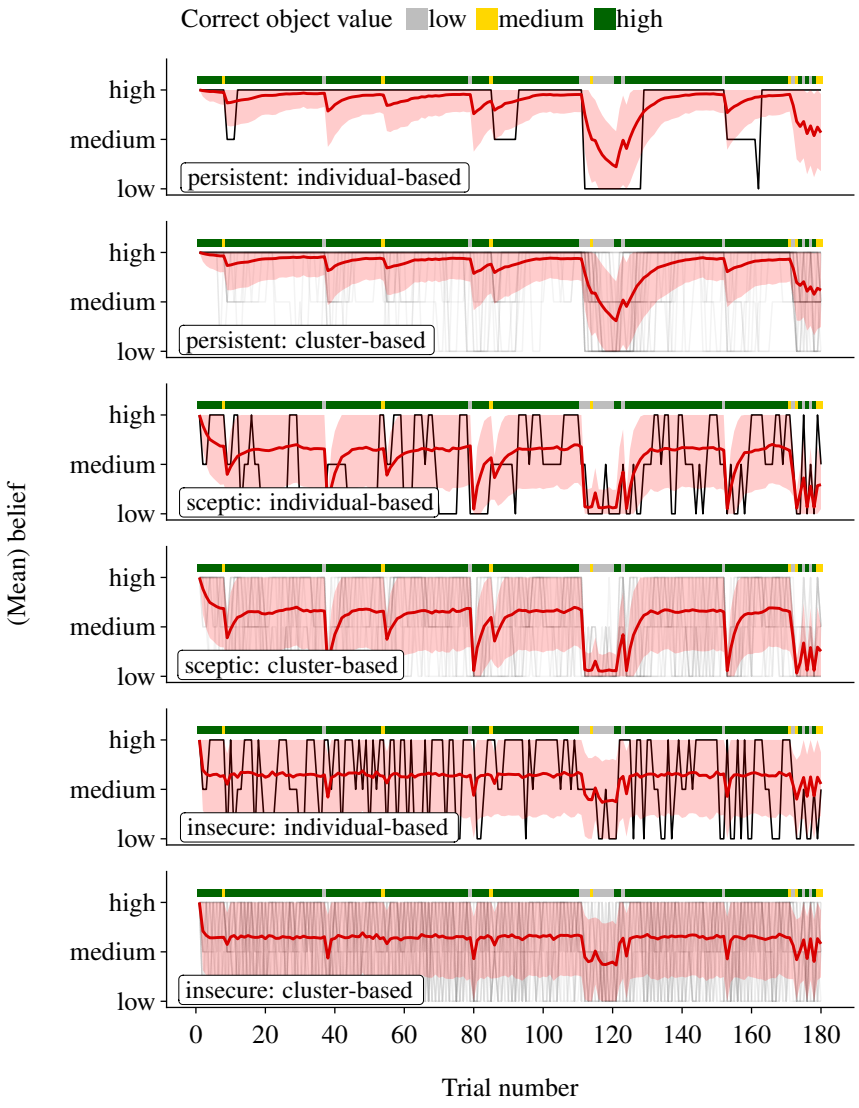


Figure 5. Mean predictions of the belief for each trial (red lines) with SD (red area) and actual participant data (black lines) are shown for the other three clusters not visualised in Fig. 4. For each cluster, an example individual-specific DBN for one participant from a cluster is presented on top of the corresponding cluster-based DBN this individual belongs to. We can see cluster 2 (here true group *persistent*) on top, cluster 4 (here true group *sceptic*) in the middle and cluster 3 (here true group *insecure*) at the bottom with colours showing the sensory evidence of each trial on top of each plot.

The six plots show the models for the three remaining clusters that have not been visualised in Fig. 4 — each once for individual-based models and for the cluster-based ones. Since the population-level model visualised in Fig. 4 represents the entire sample and is therefore not cluster-dependent, it is not shown again in Fig. 5.

The three plots for the individual-based DBNs present three different example participants that belong to three different true groups regarding their belief updating (and thus to three different clusters if they are perfectly identified). The cluster-based plots (each below the corresponding individual-based plot) visualises the predictions of the cluster-based DBNs that have been identified from the data (except for the cluster already presented before).The individuals shown in Fig. 5 belong to the cluster directly below each individual plot and are included there as one of the grey lines. Similar patterns can be identified on individual and the corresponding cluster levels although the clusters (and individuals) differ visibly regarding how easily they can be predicted, i.e., how much noise is included in their belief updating processes.

5.2 Model Comparison

We compare the models using two metrics: (a) the Brier score (Brier, 1950) and (b) the exponential mean log-likelihood. First, we calculate the mean Brier score $\overline{BS}$

$$\overline{BS} = \frac{1}{T}\sum_{t=1}^{T} BS_t, \quad BS_t = \sum_{s=1}^{S}(p_{t,s} - o_{t,s})^2$$

by taking the difference between the predicted frequency $p_{t,s}$ of each possible state s at trial t and the corresponding observation $o_{t,s}$ (either 0 or 1) which we average over all trials T. For example, let us assume we have a matrix $\mathbf{P}$ with the relative frequencies for each trial (rows) of the three possible states (columns) calculated for the 1000 iterations and an observation matrix $\mathbf{O}$, e.g.,

$$\mathbf{P} = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.2 & 0.2 & 0.6 \\ 0.3 & 0.3 & 0.4 \end{bmatrix} \quad \mathbf{O} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

meaning that, in this example, the belief was: high — high — medium — ... For the first row, these observations result in a Brier score of $(0.2-0)^2 + (0.3-0)^2 + (0.5-1)^2 = 0.38$, which can be interpreted in its range $0 \leq BS \leq 2$ with 0 being the optimal value (perfect prediction). Then, we average over all of the 1800 computed Brier scores (one score per trial).

Second, we calculate the exponential mean log-likelihood

$$\text{expMeanLL} = \exp\left(\frac{1}{T}\sum_{t=1}^{T}\ln\left(p_t(o_t) + \varepsilon\right)\right),$$

i.e., for the log-likelihood we take the logarithm of the relative frequency p_t of the actual observation o_t of each trial and average over all trials.

In this example, this is $\frac{1}{1800}\left(\ln(0.5+\varepsilon) + \ln(0.6+\varepsilon) + \ln(0.3+\varepsilon) + ...\right)$, where $\varepsilon = 1e-6$ is added to avoid $\ln(0) = -\infty$. The mean log-likelihood ($\overline{LL}$) ranges between $-\infty \leq \overline{LL} \leq 0$.

For an easier interpretability, we transform these values to $0 < e^{\overline{LL}} \leq 1$ (with 1 as optimum) by taking the exponential of the mean log-likelihood. Since we have one Brier score and exponential mean log-likelihood per individual, we average over all participants and take the SD as variation. These results are presented in Fig. 6. Better performance of the prediction made by individual and cluster DBNs compared to using the whole sample is indicated by a lower Brier score on the left and a higher value of the exponential mean log-likelihood on the right. There is no clear distinction between predictions made by individual- or cluster-based DBNs.

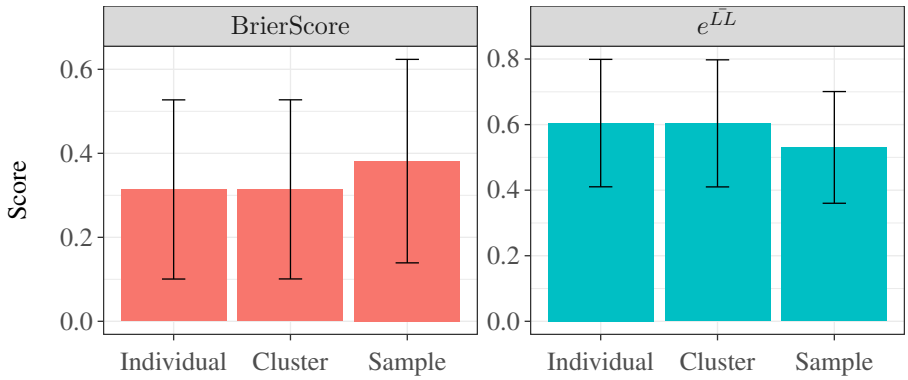


Figure 6. Presented are two plots showing the mean (coloured bars) and SD (error bars) of the Brier Score (left) and exponential mean log-likelihood $e^{\overline{LL}}$ (right). Better predictive performance is indicated by a lower Brier score on the left and a higher value of the exponential mean log-likelihood on the right.

To compare predictive performance across the three approaches (individual-level, cluster-based, and sample-based), non-parametric Friedman tests were conducted for both performance metrics across the 100 participants, as the assumptions required for parametric repeated-measures analyses were not satisfied. The analyses revealed statistically significant differences between methods for both the Brier score and the expected mean log-likelihood (see Table 1). The corresponding effect sizes indicated large effects in both cases. Post-hoc pairwise Wilcoxon signed-rank tests (Bonferroni corrected) show consistent results across both performance metrics. No statistically significant differences are observed between the cluster- and individual-based approaches, and the associated effect sizes are negligible to small. In contrast, the sample-based approach differs significantly from both the cluster-based and individual-level approaches, with large effect sizes for all comparisons.

Overall, both performance metrics led to the same conclusion: the cluster-based and individual-level approaches exhibited comparable predictive performance, whilst the sample-based approach performed significantly worse.

5.3 Comparison of Key Updating Characteristics

In addition to the general predictive performance, we also present specific aspects that we consider important from a theoretical perspective when we are interested in belief updating in such a context:

Metric	Comparison/Test	Statistic	p_{adj}	Effect Size
Friedman tests				
Brier score	Friedman $\chi^2(2)$	121.52	$< 2.2 \times 10^{-16}$	$W = 0.608$
$\exp\left(\bar{LL}\right)$	Friedman $\chi^2(2)$	116.34	$< 2.2 \times 10^{-16}$	$W = 0.582$
Post-hoc Wilcoxon signed-rank tests				
Brier score	Cluster vs. Individual	–	1.00	$r = 0.021$
Brier score	Cluster vs. Sample	–	3.51×10^{-17}	$r = 0.856$
Brier score	Individual vs. Sample	–	2.92×10^{-17}	$r = 0.858$
$\exp\left(\bar{LL}\right)$	Cluster vs. Individual	–	0.825	$r = 0.109$
$\exp\left(\bar{LL}\right)$	Cluster vs. Sample	–	$< 2.2 \times 10^{-16}$	$r = 0.860$
$\exp\left(\bar{LL}\right)$	Individual vs. Sample	–	$< 2.2 \times 10^{-16}$	$r = 0.858$

Table 1. For non-parametric comparisons of predictive performance across methods, degrees of freedom for the Friedman test are reported in brackets with effect size Kendall W and rank-biserial effect size r for the Wilcoxon signed-rank tests.

1. Frequency of each belief state, i.e., person A beliefs that person B chooses a (a) high-, (b) medium-, or (c) low-value object.

2. The probability that the belief is updated:

   (a) $p\left(b_t \neq b_{t-1}\right)$: the general probability that the belief b_t changes in one trial compared to the previous b_{t-1} (relative frequency of belief updates)

   (b) $p\left(b_t \neq b_{t-1} \mid e_t \neq e_{t-1}\right)$: the probability that the belief b_t changes in one trial compared to the previous one b_{t-1} if the sensory evidence e_t has also *changed* compared to the previous trial e_{t-1}

   (c) $p\left(b_t \neq b_{t-1} \mid e_t = e_{t-1}\right)$: the probability that the belief b_t changes in one trial compared to the previous one b_{t-1} if the sensory evidence e_t has *remained the same* compared to the previous trial e_{t-1}

For each participant, we calculate the mean estimate of these values over the 1000 iterations — either from the individually fitted, cluster-based or whole sample DBNs — and compare them to the actual values in the data of that participant to get an error and a SD of these estimates over all iterations. These error values are shown summarised for all clusters together in Fig. 7. Since the pattern is similar and we are mainly interested in the general performance — independent of the exact cluster — we only report the average over the clusters here. The mean error of all estimates is much higher when predictions are made based on the DBN fitted on the whole sample compared to the other two modelling approaches. However, the performance does not seem to be worse if cluster-based DBNs are used compared to individual-specific models. There are also some differences in the quality of the estimates between the four clusters.

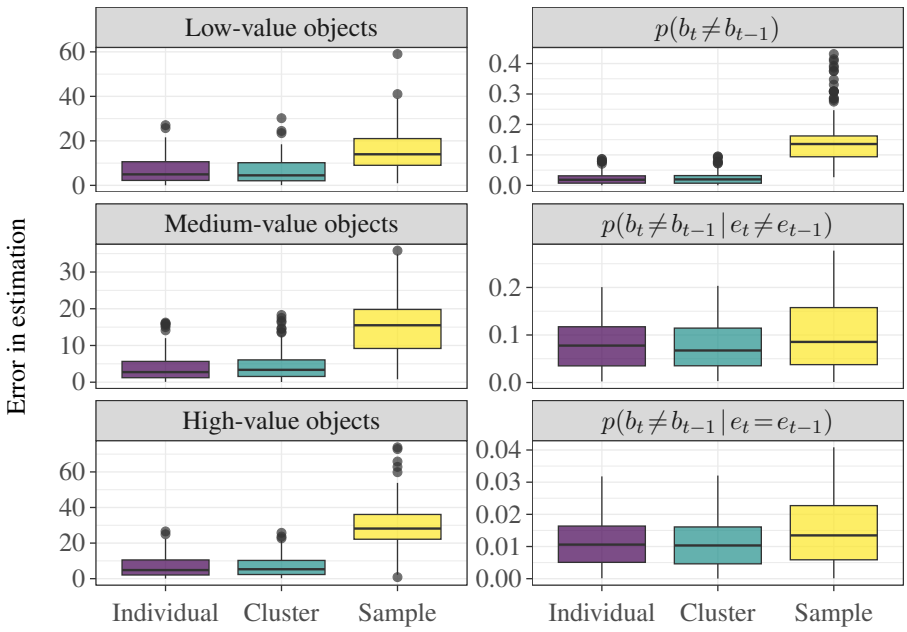


Figure 7. The plots visualise different errors in estimation by the three modelling approaches. On the left, errors (as counts) in the estimation of the three different belief states — low (top), medium (middle), and high (bottom) — and on the right, errors in estimating the relative frequency of changes in beliefs in general (top), after evidence has changed (middle) or remained the same (bottom) are shown.

5.4 Validation With Less Learning Trials

Since the model had a lot of learning trials, we repeat the same analysis but with less trials in the fitting process. In the real existing experimental study, there are 60 trials per block. Here, we repeat each of the three blocks only twice, ending up with 360 trials that are used for learning the DBNs. This increases the intra-cluster variability of individuals, thereby decreasing distances between some individuals of different groups, which in turn makes these individuals harder to distinguish and cluster to one of the groups. Nevertheless, the results show equivalent patterns. The main difficulty seems to be the correct identification of the clusters, which still seems to work sufficiently well with 360 trials. However, due to the complexity of the study setup, further reducing the number of trials, e.g., to 180, leads to partly undistinguishable groups, which can result in inconsistent clusters and especially a difficult identification of the correct number of clusters. A systematic investigation of how the minimum number of trials should be chosen depending on the concrete setting (number of variables and groups, structure of trials, etc.) and how robust the predictions are for imperfectly identified clusters is necessary but beyond the scope of this paper.

6. Discussion

We have compared individual-specific models with cluster-based and population-level ones regarding their predictive accuracy of belief updating as a cognitive process. The results indicate that clustering-based modelling provides predictions that are on average as accurate as predictions based on models fitted specifically on the individual participants. Both these approaches (individual- and cluster-based) outperform models fitted on the full sample. In particular, cluster-derived predictions appear to predict meaningful structure in belief updating processes just as good as individual-based models. This supports the idea that latent subgroups, when properly identified, offer a useful compromise between overly noisy and not generalisable individual estimates and overly averaged sample-level models. Importantly, such clustering approaches may also facilitate prediction for new participants, provided that their characteristics align sufficiently with those of the identified groups. In principle, group membership could be inferred from independent pre-test measures such as questionnaires or other behavioural assessments (which would require empirical validation in corresponding studies). If reliable mappings between such measures and cluster assignment (i.e., process of belief updating) can be established, this would enable predictions for new individuals regarding their belief updating processes without the need to run the full experimental task, thereby extending the practical applicability of the model to settings where only pre-existing participant information is available. The results have been demonstrated on data simulated from a realistic experimental context including complexity from the inclusion of other noisy variables, as well as from the fact that not all trial transitions and stimulus conditions were equally represented, which may have constrained learning and biased parameter estimation but represent realistic limitations for applications to real data. If the different trials are more equally represented (i.e., in our setting that there are equally many trials with sensory evidence *low*, *medium*, and *high*) the estimation of the CPTs should be even more reliable, especially for a smaller number of trials, since the frequencies of all possible combinations of the different states are more balanced in this case and less trials are needed to estimate all various conditional probabilities.

Introducing a social-cognitive perspective serves here primarily to contextualise the clustering methodology rather than to modify the clustering process itself. By treating belief updating as an instance of cognitive behaviour, the proposed approach can be considered in relation to phenomena that extend beyond the purely computational representation of dynamic Bayesian networks. Such an application illustrates how patterns detected algorithmically may subsequently be interpreted as meaningful differences in processes that occur in natural settings. At the same time, this rationale is not specific to belief updating. The same framework could be applied to a range of other cognitive or behavioural processes, as long as these processes exhibit both meaningful individual differences and sufficient commonality to support the formation of coherent groups. The suitability of a particular fitting strategy may furthermore depend on the structure of this heterogeneity: the approach may differ in its usefulness depending on how much individuals within a given cluster vary from one another and how pronounced the differences are between clusters across the population.

Some limitations should be considered. The validity of the clustering approach depends strongly on the assumption that the underlying group structure is well captured, which may not always hold. If clustering is imperfect, results may be distorted which potentially reduces generalisability.

Future work should focus on further validating the robustness of these findings, for example through additional variations across different datasets, the inclusion of unobservable variables, and simulations that systematically vary the number of trials, structure and group separability. In particular, the simulations considered a relatively large number of trials per participant, raising the question of whether lower trial numbers might reveal conditions under which cluster-level fitting even provides an advantage over individual-level fitting. When data is sparse at the individual level, there may be insufficient information to reliably estimate individual-specific parameters, while the data within a cluster may still provide sufficient information for reliable parameter estimation. The same consideration may apply to population-level fitting, where clustering data across individuals could potentially compensate for the limited amount of data available for each participant. However, finding accurate clusters with as few trials might be challenging as it increases the inter-cluster variability which can lead to a reduced distinguishability. Systematically varying the number of trials per participant could therefore help identify whether and under which conditions problems or benefits of models at the cluster or population level emerge. Furthermore, different groups were only implemented in one variable. Including groups in the other variables — that might be equivalent or different to the one in belief — brings additional complexity and would need further considerations about how to best make predictions based on cluster-based models.

7. Conclusion

Clustering individuals via hierarchical agglomerative clustering on their CPTs can identify groups with similar belief updating patterns. Fitting one DBN per cluster yields predictions that are, on average, as accurate as those of individual-specific DBNs and substantially better than predictions of a single DBN estimated from the whole sample. Both evaluated performance metrics — the Brier score and the exponential mean log-likelihood — as well as additional belief updating characteristics consistently support this conclusion. Cluster-based models therefore offer a principled compromise that captures meaningful subgroup-specific patterns of individual cognitive processes such as the way people update or maintain their beliefs while allowing interpretation of the provided clusters and retaining the generalisability of population-level models, which makes them particularly attractive when predictions of cognitive processes should extend to previously unseen individuals.

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